Experiment - 5

ANALYSIS OF DATA COMMUNICATION IN A WEB APPLICATION USING BROWSER DEVELOPER TOOLS

AIM

To analyze the data communication between a web browser and a web server using the **Network tab of Browser Developer Tools**

OBJECTIVES

- To monitor network requests generated by a web application.
- To identify HTTP methods such as GET and POST.
- To analyze request and response headers.
- To inspect request payload and response data.
- To study HTTP status codes.
- To analyze data size and loading time.
- To identify network errors and performance issues

PROCEDURE

1. Open Google Chrome and launch a web application.
2. Right-click on the webpage and select **Inspect**.
3. Select the **Network** tab.
4. Clear the existing network records.
5. Reload the webpage.
6. Observe the requests generated by the browser.
7. Select any request from the Network list.
8. Examine its URL, method, status, size, and time.
9. Open the **Headers** section and examine request and response headers.
10. Perform an action such as searching or submitting a form.
11. Observe the new request generated by the action.
12. Record the important observations

NETWORK REQUEST ANALYSIS

When a webpage is loaded, the browser sends several requests to the server. These requests may contain HTML, CSS, JavaScript, images, fonts, and API data.

S.No	Resource	Method	Status	Type
1	index.html	GET	200	Document
2	style.css	GET	200	Stylesheet
3	script.js	GET	200	Script
4	api/users	GET	200	Fetch
5	api/login	POST	200	Fetch

Observation: The browser uses different requests to obtain webpage resources and communicate with the server.

HTTP METHOD ANALYSIS

GET

Used to retrieve information from the server.

POST

Used to send data to the server.

PUT

Used to update existing information.

DELETE

Used to delete a resource.

REQUEST AND RESPONSE ANALYSIS

After selecting a request, the following information can be observed.

Request Information

- Request URL
- Request Method
- Request Headers
- Query Parameters
- Payload

Response Information

- Status Code
- Response Headers
- Response Data
- Content Type
- Response Size

The browser sends a request to the server, and the server returns the required data as a response.

PAYLOAD AND RESPONSE ANALYSIS

The **Payload** section displays information sent from the browser to the server.

The **Response** section displays information returned by the server.

Example

This allows developers to verify whether the correct data is being transmitted and received.

PERFORMANCE ANALYSIS

The Network tab also provides information about the performance of each request.

Important parameters include:

- Request time
- Response time
- File size
- Number of requests
- Waiting time
- Download time

ERROR ANALYSIS

The Network tab helps identify failed requests. This indicates that the requested resource was not found. This indicates an authentication or authorization problem. Server-side problems may appear as: Thus, status codes and response messages help developers locate communication problems.

RESULT

The data communication of a web application was successfully analyzed using Browser Developer Tools.

The Network tab was used to observe:

- HTTP requests
- HTTP methods
- Request and response headers
- Payload
- Response data
- Status codes
- Data size
- Loading time
- Network errors

CONCLUSION

Browser Developer Tools provide an effective way to monitor communication between a web browser and web server. The Network tab helps developers understand requests and responses, identify errors, and analyze the performance of web applications.